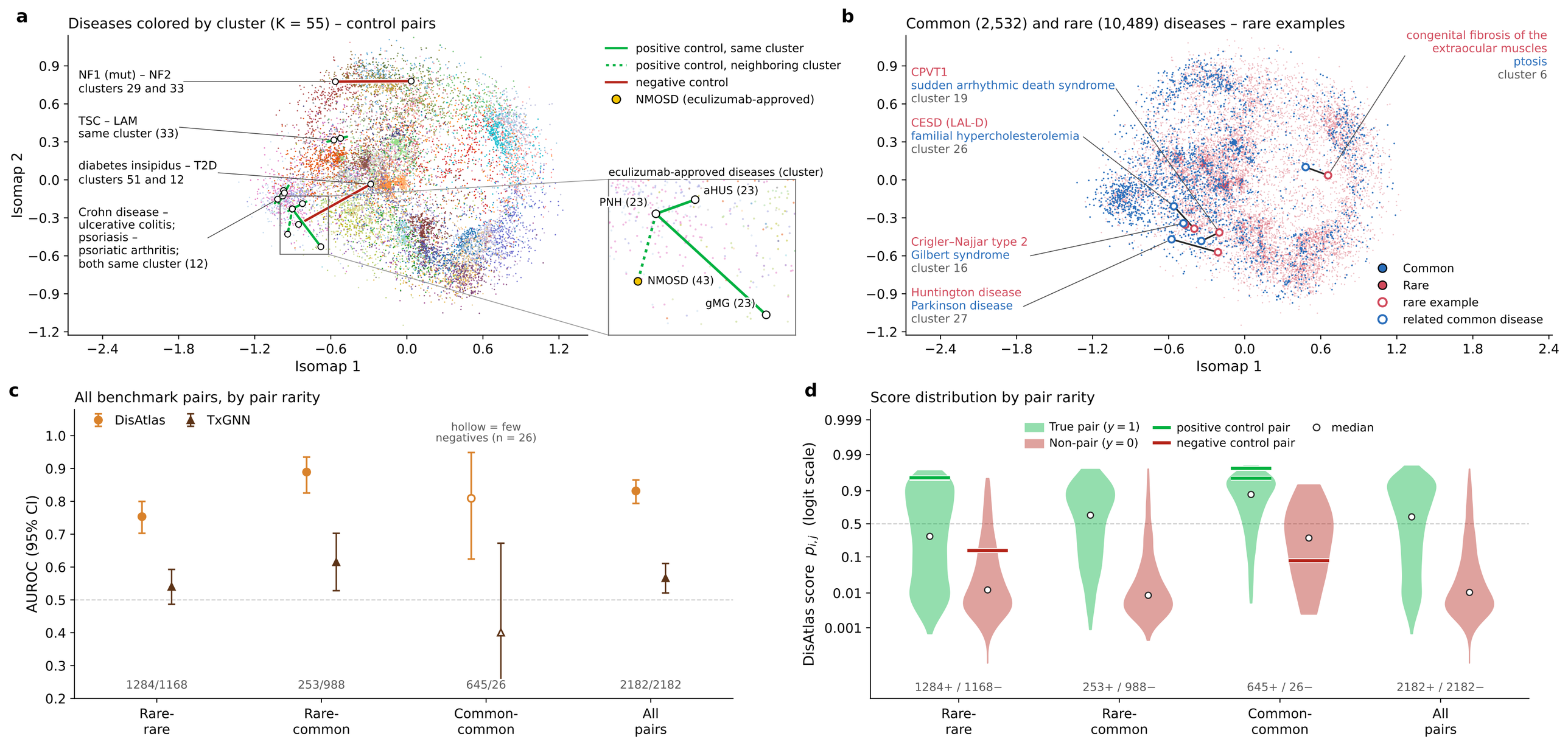

a
Diseases colored by cluster (K = 55) – control pairs
positive control, same cluster
positive control, neighboring cluster
negative control
NMOSD (eculizumab-approved)
NF1 (mut) – NF2
clusters 29 and 33
TSC – LAM
same cluster (33)
diabetes insipidus – T2D
clusters 51 and 12
Crohn disease –
ulcerative colitis;
psoriasis –
psoriatic arthritis;
both same cluster (12)
eculizumab-approved diseases (cluster)
aHUS (23)
PNH (23)
NMOSD (43)
gMG (23)
Isomap 1
Isomap 2
b
Common (2,532) and rare (10,489) diseases – rare examples
congenital fibrosis of the
extraocular muscles
ptosis
cluster 6
CPVT1
sudden arrhythmic death syndrome
cluster 19
CESD (LAL-D)
familial hypercholesterolemia
cluster 26
Crigler–Najjar type 2
Gilbert syndrome
cluster 16
Huntington disease
Parkinson disease
cluster 27
Common
Rare
rare example
related common disease
Isomap 1
c
All benchmark pairs, by pair rarity
DisAtlas
TxGNN
hollow = few
negatives (n = 26)
AUROC (95% CI)
1284/1168
253/988
645/26
2182/2182
Rare-
rare
Rare-
common
Common-
common
All
pairs
d
Score distribution by pair rarity
True pair (y = 1)
Non-pair (y = 0)
positive control pair
negative control pair
median
DisAtlas score $p_{i,j}$ (logit scale)
1284+ / 1168−
253+ / 988−
645+ / 26−
2182+ / 2182−
Rare-
rare
Rare-
common
Common-
common
All
pairs